Building the Community Library Network in Filius

COM4401 Networking Fundamentals: practice task (individual)

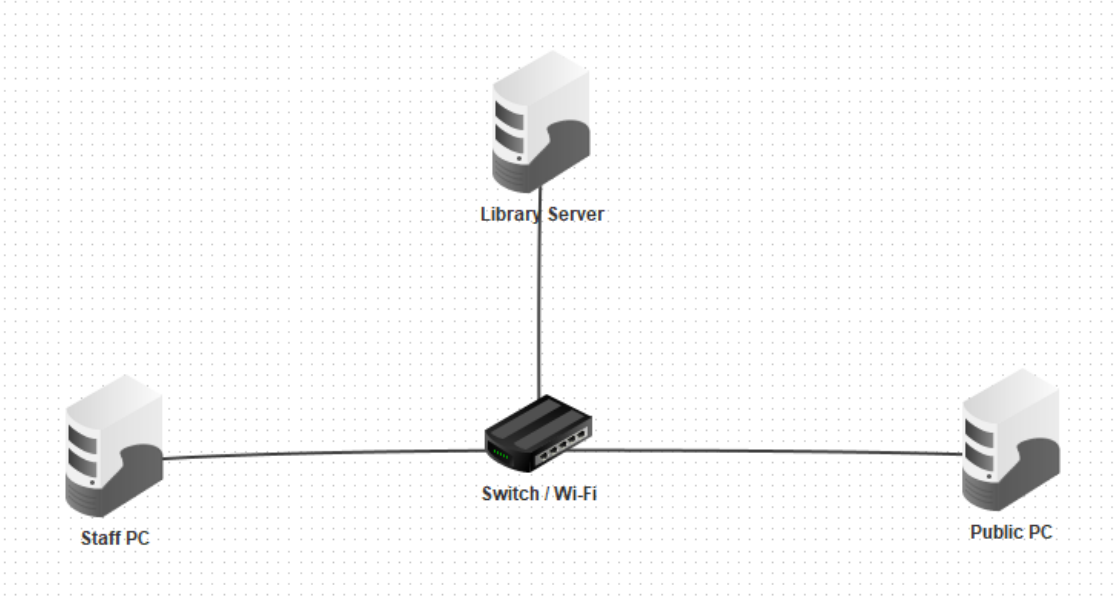
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Part 1: The network

Screenshot 1: the network

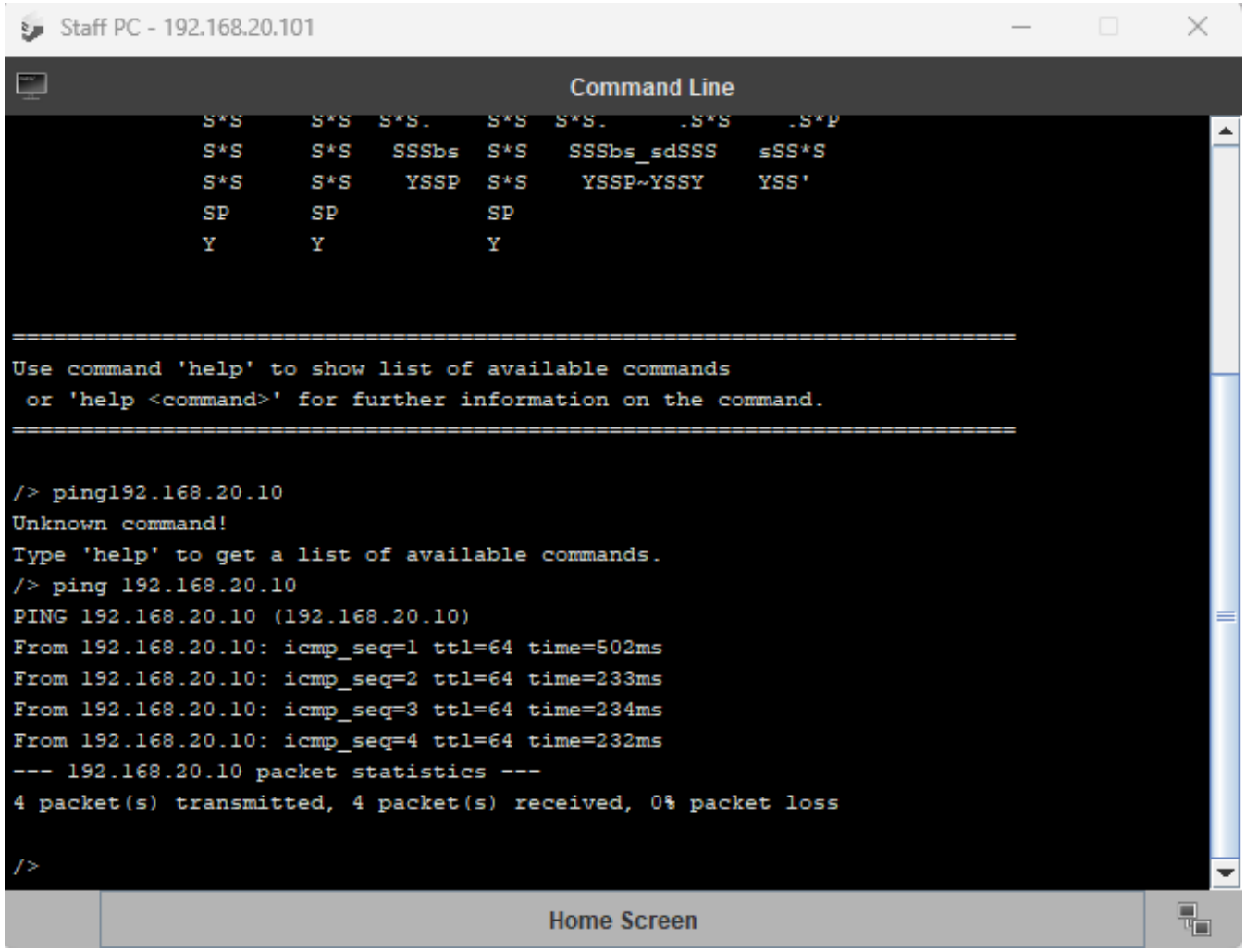
Three named computers and one switch, all cabled together in Design mode.



Screenshot 1: Staff PC, Public PC and Library Server, each cabled to the switch.

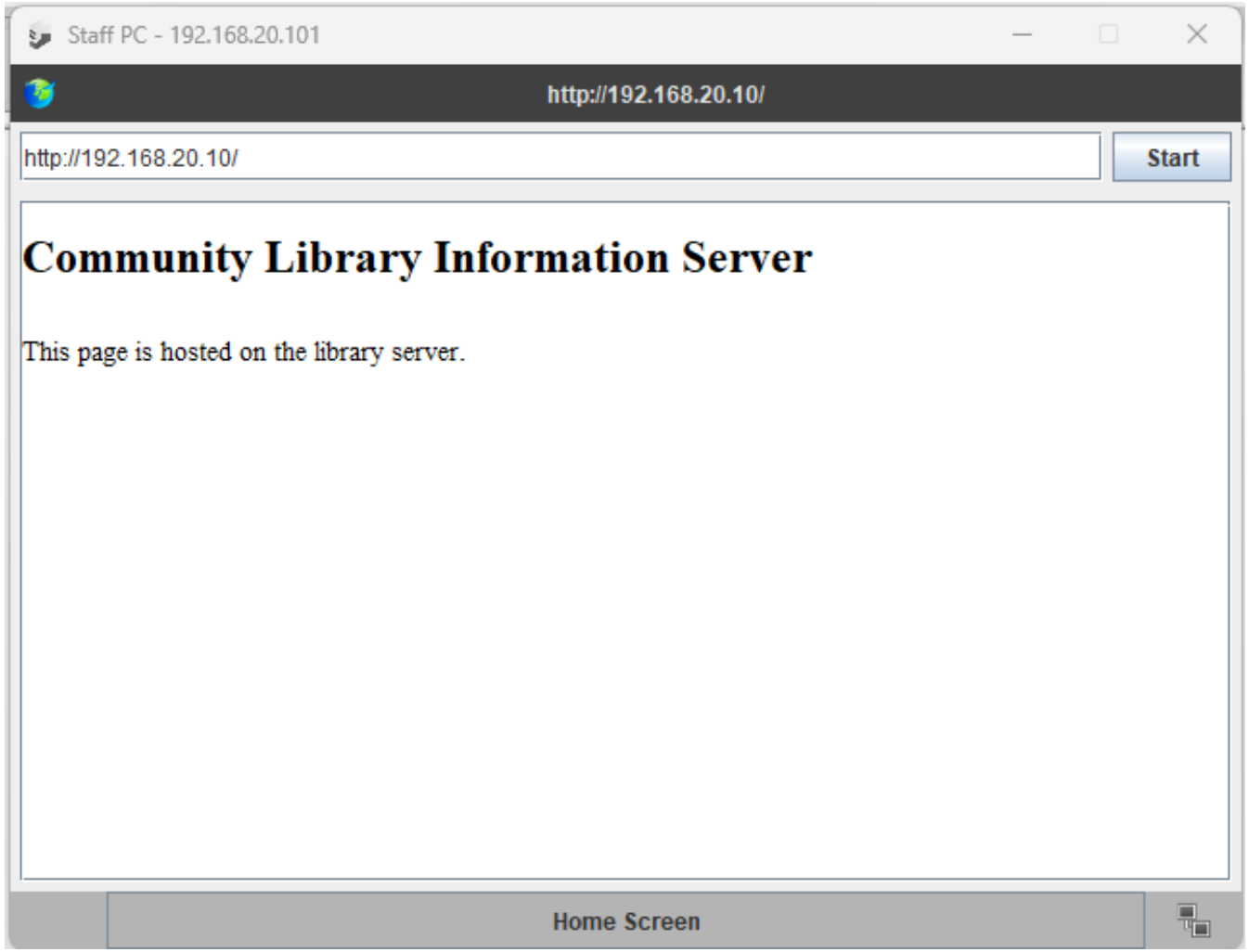
Device	IP address	Subnet mask	Gateway	DNS
Library Server	192.168.20.10	255.255.255.0	blank	blank
Staff PC	192.168.20.101	255.255.255.0	blank	blank
Public PC	192.168.20.102	255.255.255.0	blank	blank

Screenshot 2: ping from the Staff PC



Screenshot 2: the Staff PC's Command Line showing 4 replies from 192.168.20.10 with 0% packet loss.

Screenshot 3: the library page in the Staff PC’s browser



Screenshot 3: the Staff PC’s web browser showing the Community Library Information Server page from http://192.168.20.10.

Part 2: Client and server

In our network, the Staff PC was the client and the Library Server was the server. The Staff PC asked for the website and the Library Server provided it. First the Staff PC pinged the server at 192.168.20.10 to check it could reach it, and got 4 replies back with 0% packet loss. Then it opened http://192.168.20.10 in its browser, and the server sent back index.html, the page I saved on it.

Part 3: Protocols and layers

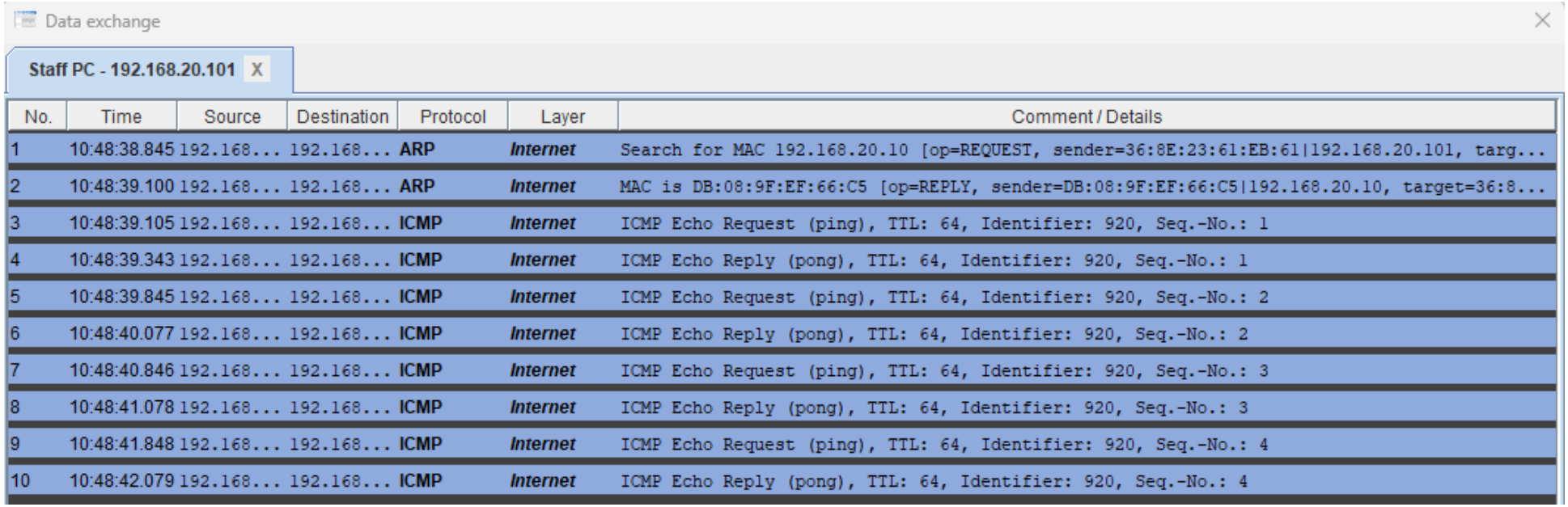
Evidence table

Protocol	Screenshot	Top layer reached	What it did
ARP	Figure A1 and A2	Internet	The Staff PC asked every device “who has 192.168.20.10?” and the Library Server replied with its MAC address.
ICMP	Figure A1 and A3	Internet	The ping: 4 echo requests (“are you there?”) and 4 echo replies.
TCP	Figure B1 and B2	Transport	The 3 handshakes (SYN, SYN/ACK, ACK) set up the connection before the page was sent.
HTTP	Figure B1 and B3	Application	The Staff PC asked for the page (GET) and the server sent it back (200 OK).
DNS (extension)	Not done		Optional extension, not attempted in this practice.

Why did ping use fewer layers than loading the web page?

The ping was just a quick knock on the door, so ICMP only used the Network and Internet layers to ask “are you there?” and get a reply. The web page needed more of an interaction: the Staff PC and the server did the 3 handshakes with TCP on the Transport layer (SYN, SYN/ACK, ACK), then HTTP carried the request and the page on the Application layer. In Filius the HTTP row had all four layers filled in, each one wrapped inside the one below, while the ping rows only had two.

Figures



No.	Time	Source	Destination	Protocol	Layer	Comment / Details
1	10:48:38.845	192.168...	192.168...	ARP	*Internet*	Search for MAC 192.168.20.10 [op=REQUEST, sender=36:8E:23:61:EB:61|192.168.20.101, targ...
2	10:48:39.100	192.168...	192.168...	ARP	*Internet*	MAC is DB:08:9F:EF:66:C5 [op=REPLY, sender=DB:08:9F:EF:66:C5|192.168.20.10, target=36:8...
3	10:48:39.105	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 1
4	10:48:39.343	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 1
5	10:48:39.845	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 2
6	10:48:40.077	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 2
7	10:48:40.846	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 3
8	10:48:41.078	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 3
9	10:48:41.848	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 4
10	10:48:42.079	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 4

Figure A1: Data Exchange on the Staff PC during the ping. Rows 1 and 2 are ARP, rows 3 to 10 are ICMP.

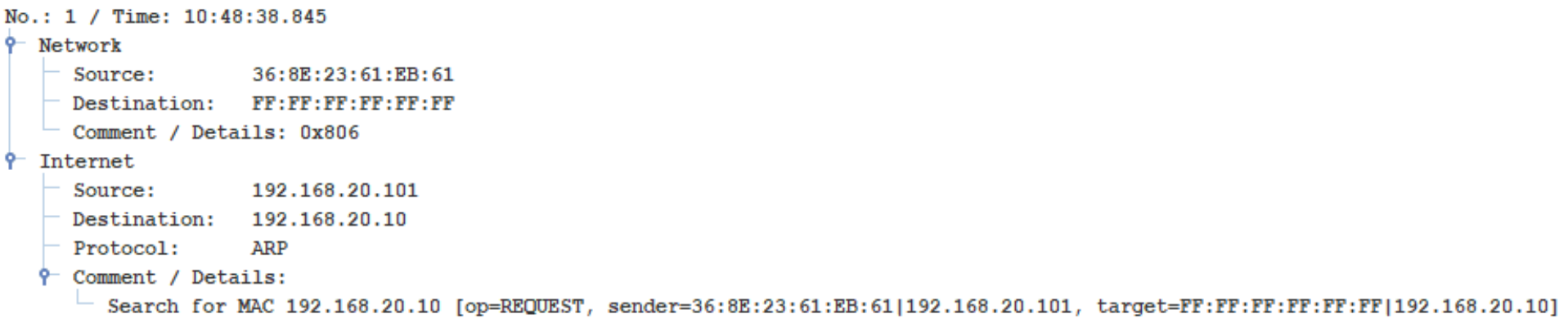


Figure A2: ARP row opened. Only the Network and Internet layers are filled in.

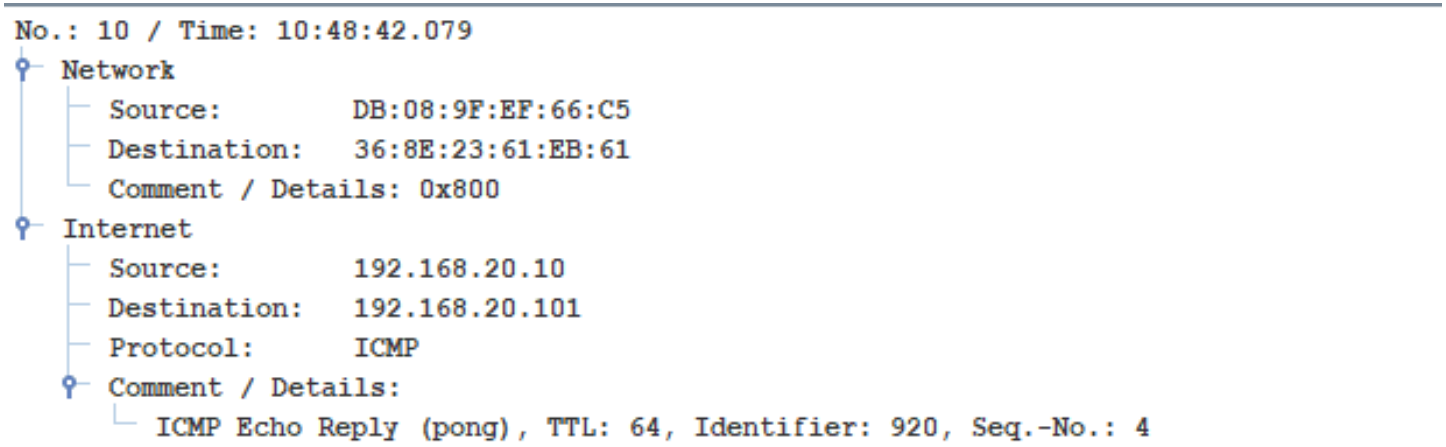


Figure A3: ICMP row opened (echo reply). Only the Network and Internet layers are filled in.

Staff PC - 192.168.20.101 X

No.	Time	Source	Destination	Protocol	Layer	Comment / Details
3	10:48:39.105	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 1
4	10:48:39.343	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 1
5	10:48:39.845	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 2
6	10:48:40.077	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 2
7	10:48:40.846	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 3
8	10:48:41.078	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 3
9	10:48:41.848	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Request (ping), TTL: 64, Identifier: 920, Seq.-No.: 4
10	10:48:42.079	192.168...	192.168...	ICMP	*Internet*	ICMP Echo Reply (pong), TTL: 64, Identifier: 920, Seq.-No.: 4
11	11:07:37.857	192.168...	192.168...	TCP	*Transport*	SYN, SEQ: 2,000,000
12	11:07:38.089	192.168...	192.168...	TCP	*Transport*	SYN, SEQ: 1,000,000, ACK: 2,000,001
13	11:07:38.098	192.168...	192.168...	TCP	*Transport*	SEQ: 2,000,001, ACK: 1,000,001
14	11:07:38.170	192.168...	192.168...	HTTP	*Application*	GET / HTTP/1.1 Host: 192.168.20.10
15	11:07:38.399	192.168...	192.168...	TCP	*Transport*	SEQ: 1,000,001, ACK: 2,000,036
16	11:07:38.456	192.168...	192.168...	HTTP	*Application*	HTTP/1.1 200 OK Content-type: text/html <html> <body> <h1>Community Library In...
17	11:07:38.465	192.168...	192.168...	TCP	*Transport*	SEQ: 2,000,036, ACK: 1,000,179
18	11:07:38.524	192.168...	192.168...	TCP	*Transport*	FIN, SEQ: 2,000,036
19	11:07:38.759	192.168...	192.168...	TCP	*Transport*	SEQ: 1,000,179, ACK: 2,000,037
20	11:07:38.817	192.168...	192.168...	TCP	*Transport*	FIN, SEQ: 1,000,179
21	11:07:38.826	192.168...	192.168...	TCP	*Transport*	SEQ: 2,000,037, ACK: 1,000,180

Figure B1: Data Exchange on the Staff PC while loading the page. Rows 11 to 13 are the TCP handshake, rows 14 and 16 are HTTP.

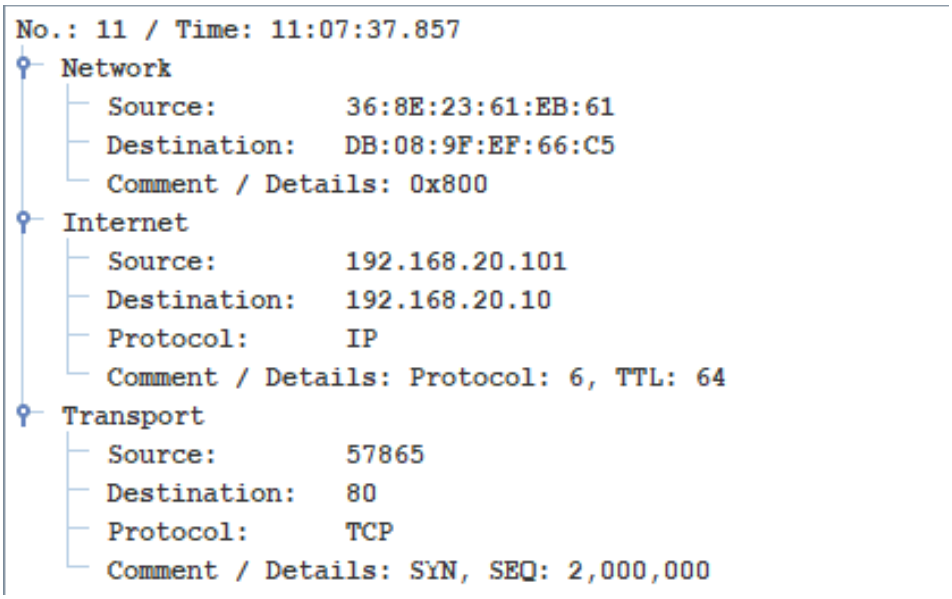


Figure B2: TCP row opened (SYN). The Network, Internet and Transport layers are filled in.

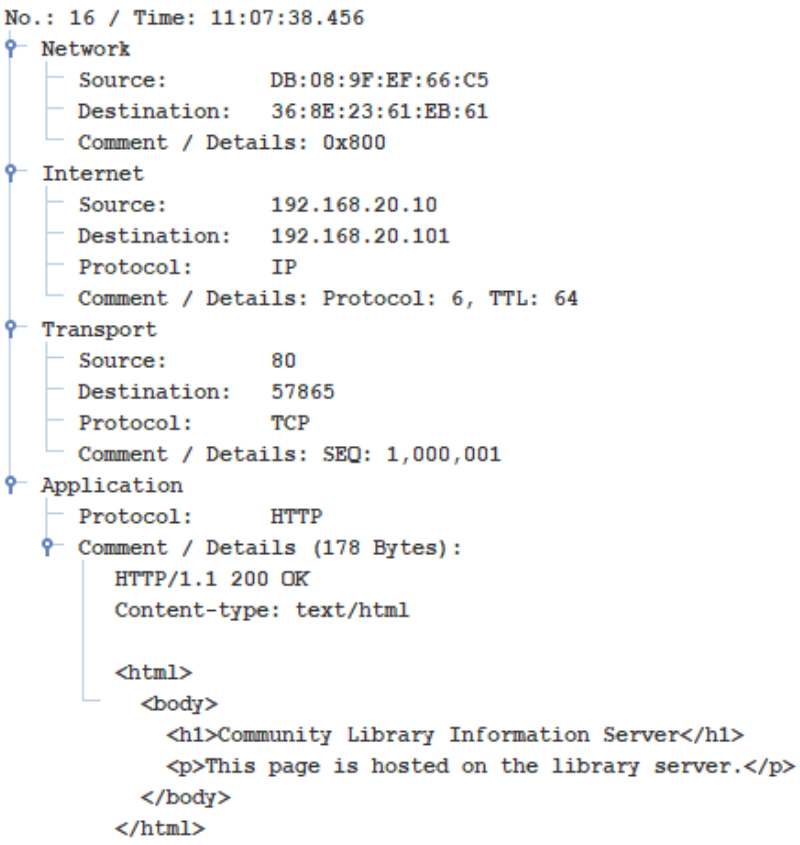


Figure B3: HTTP row opened (200 OK). All four layers are filled in, with the page inside.